

MGMT 264: Venture Capital and Entrepreneurial Management

Fundraising PPT Supplement: Calculations relating to convertible note conversions

Calculations to support slide #13 (Conversion on a Post Money basis):

Technically, calculating the note into a post-money equity position is what the language of the note calls for.

Step 1: calculate the price per share of the conversion of the Note

$$\$5.00 \times .8 = \$4.00/\text{share}$$

Step 2: Calculate the number of shares

$$\$500,000 / \$4.0/\text{share} = 125,000 \text{ shares}$$

Note: The accumulated interest is usually added to the face value of the note in this step.

Step 3: calculate the % of ownership of convertible note holders

- Series A: 1,000,000 shares (negotiated \$5/sh for \$5M) = 32%
- Founders: 2,000,000 shares = 64%
- Notes: 125,000 shares = 4%

Total number of shares after conversion = 3,125,000

Calculations to support slide #15 (Conversion on a Pre-Money basis)

Because the Series A objects to receiving less than their negotiated 33% stake in the company, we need to bake the note into the pre-money valuation. (This can also cause some resistance from the founders, since including the notes reduces the pre-money of the company, even though the debt is really just a bookmark for an equity security.)

[pre-money value – note amount / (1 - discount)] / fully diluted capital = Series A price/share

$$[\$10\text{M} - \$500\text{K} / .80] / 2\text{M} = \$4.69/\text{Series A share}; \text{discounted conversion price} = \$3.75/\text{share}$$

Step 1: Establish the adjusted pre-money valuation of the company.

- \$10M subtract the “leveraged” value of the debt, i.e., the face amount of the debt / (1 - discount) = \$500K / .80 = \$625K.
- So the pre-money is \$10M - \$625K = \$9.375M

Step 2: Calculate the Series A price / share.

- Adjusted pre-money valuation / fully diluted capitalization
- \$9.375M / 2m = \$4.69/ Series A share
- \$5M / \$4.69 = 1,066,667 shares

Step 3: Calculate the note conversion price and the number of conversion shares

- \$4.69 x .8 = \$3.75 per conversion share
- \$500K/\$3.75 = 133,333 conversion shares

Step 4: Equity ownership

- Founders: 2,000,000 shares = 62.5%
- Series A: 1,066,667 shares = 33.3%
- Notes: 133,333 shares = 4.2%

Comment to slide #16:

The explanation on the purpose of a valuation cap.

Example: a note investor puts up \$1M into a seed financing, and is expecting that the company will raise Series A financing at a pre-money valuation of about \$4m. Absent discount conversion prices and other bells and whistles in the terms of the note, the note investor on conversion would end up with a 20% ownership in the company (\$1m against a \$5m post-money valuation). But what if the Series A financing comes in at an \$11 million pre-money valuation (i.e., the average pre-money valuation for the industry for 2018). Again absent discounts, etc., now the note investor ends up on conversion with \$1m/\$12 million ownership, or 8.3%. This kills a lot of the incentive for the note investor to put \$ into the company at all. The valuation cap is there to give the note investor protection against this “lightning in a bottle” situation. This protection is there as an alternative to the discounted conversion price usually contained in a note, where the conversion discount doesn’t protect the investor enough.

Calculations to support slide #18 (Valuation Cap & Discount: Conversion on a Pre-Money Basis)

Step 1: Calculate price per conversion share—does discount OR cap result in the lower price for the notes?

- Discount price calculation, pre-money: \$3.75 (see prior slide)

OR

- Cap price calculation, pre-money:
 - Pre-money valuation as capped / fully diluted capital
 - \$4M / 2m = \$2.00 per conversion share
- So use \$2.00 per conversion share

Step 2: Calculate number of conversion shares

- \$500K / \$2 = 250,000 conversion shares

Sep 3: Calculate Series A price

- Pre-money valuation / fully diluted capitalization
- Fully diluted # of shares = 2,000,000 + 250,000 = 2,250,000
- \$10M / (2m shares of common + 250K conversion shares) (remember, the conversion shares are going into the pre-money capitalization) = **\$4.44/series A shares**

Step 4: Calculate # of Series A Shares

Investment/price per share

$$\$5,000,000 / \$4.444444 = 1,125,000$$

- Pre-money valuation / fully diluted capitalization
- \$10M / 2m + 250K (remember, the conversion shares are going into the pre-money)
- \$4.44 per Series A share

Step 4: Calculate ownership

- Founders: 2,000,000 = 59.3%
- Series A: 1,125,000 = 33.3%
- Notes: 250,000 = 7.4%

Calculations to support slide #19

Note, there is a common problem here, as with all note conversions where there is either a discount or a cap. It is as follows:

- Investors generally don't like the notes getting preferred shares with a liquidation preference that exceeds the note amount being converted. Using the above example, the notes are getting 250,000 shares of Series A that automatically have an aggregate liquidation preference of \$1,111,111—but the cancellation of debt is only \$500,000 (again ignoring interest). So the notes are getting Series A shares with an excess liquidation preference of \$611,111! In effect, this enables the note holders to get a better arrangement than the Series A investors who are getting shares with a liquidation preference that is equal to the cash value of their investment.
- To correct for this, in most instances, the conversion arrangement frequently will require for Series A conversion shares equal to $\$500,000 / \$4.44 = 112,500$ shares of Series A. The balance of the 137,500 shares remaining from the 250,000 shares owing to the note holders will then be issued as common stock.